

BlotQuant: A Browser-Based Platform for Automated Western Blot Quantification, Uncertainty Estimation, and Authenticity Verification

Author Name

Department of Computer Science and Engineering

University Name, City, Country

email@university.edu

Abstract — Western blot analysis is a foundational technique in molecular biology, yet its quantitative interpretation remains heavily dependent on manual, operator-biased workflows. Existing computational tools treat band detection, background subtraction, and normalization as independent manual steps, and none provide measurement uncertainty estimates or assess image authenticity. This paper presents BlotQuant, an open-source, browser-based platform that automates the complete western blot analysis pipeline. The system accepts raw blot images and produces structured outputs comprising band localization, relative intensity values with 95% confidence intervals, normalization strategy inference, and an authenticity score indicating the probability that the image represents genuine experimental data. The current implementation employs a multimodal vision language model (Gemini 1.5 Flash) through structured prompting to serve all five analytical modules, while the architecture supports drop-in replacement with dedicated deep learning models as training data becomes available. The platform is built on FastAPI with a React frontend and MongoDB persistence. Preliminary validation demonstrates consistent automated analysis completing in under twenty seconds per image, compared to several minutes of manual densitometry with measurable inter-operator variability.

Keywords — western blot quantification; band detection; uncertainty estimation; image authenticity; vision language models; bioimage analysis

I. INTRODUCTION

The western blot assay, first described by Towbin et al. in 1979 [1], remains among the most widely used techniques in biological and medical research for detecting and semi-quantifying specific proteins. The assay produces an image in which protein bands appear at positions corresponding to their molecular weight, and the optical density of each band is assumed to correlate with protein abundance under appropriate experimental conditions.

Despite the maturity of the wet-lab protocol, the computational analysis of western blot images has not kept pace. The predominant workflow still involves manual region-of-interest selection in tools such as ImageJ [2], subjective background subtraction, researcher-chosen normalization strategies, and export of raw intensity values with no associated uncertainty. This workflow suffers from three well-documented shortcomings: inter-operator variability of up to 12% for the same image [3], absence of measurement confidence intervals, and no mechanism for detecting fabricated or manipulated images.

The Journal of Biological Chemistry has explicitly identified normalization strategy selection as one of three major reporting gaps in published western blot data [4]. Simultaneously, the emergence of generative AI models capable of producing photorealistic blot images has introduced an urgent need for purpose-built authenticity verification tools [5].

This paper presents BlotQuant, a platform that addresses these gaps through an integrated, automated pipeline requiring no manual parameter selection. The system accepts a raw blot image and produces five structured outputs: band localization coordinates, relative intensity measurements with 95% confidence intervals, inferred normalization strategy, uncertainty estimates, and an authenticity assessment score. The following sections describe the system architecture, implementation methodology, validation approach, and results.

II. RELATED WORK

A. Manual and Semi-Automated Densitometry

ImageJ's gel analysis module [2] provides the most widely used free densitometry workflow. Users manually define lane boundaries, generate one-dimensional intensity profiles, and integrate peaks using the trapezoidal method. The process is entirely dependent on operator judgment at every stage. Commercial alternatives from Bio-Rad (Image Lab) [6] and LI-COR (Image Studio) [7] offer more polished interfaces but maintain the same fundamental requirement for manual

methodological decisions and provide no uncertainty quantification.

B. Deep Learning Applied to Blot Images

Convolutional neural networks have achieved substantial success in biomedical imaging tasks including cell segmentation, histopathology, and radiology [8]. Their application to western blot images, however, has been limited to binary band classification rather than intensity regression [9]. This limitation stems primarily from the scarcity of publicly available blot images with ground-truth intensity annotations suitable for supervised regression training.

C. Image Authenticity Detection

Forensic detection of manipulated or synthetically generated images is an active research area [10]. However, existing tools are trained on natural photographic images and exhibit inconsistent performance when applied to the distinct visual characteristics of blot membrane images, which have specific noise signatures, texture patterns, and background gradients that differ fundamentally from photographic imagery.

III. SYSTEM ARCHITECTURE

A. Architectural Overview

BlotQuant employs a client-server architecture consisting of three layers. The presentation layer is a React single-page application providing the user interface. The application logic layer is a FastAPI backend that handles image ingestion, analysis orchestration, and result formatting. The persistence layer uses MongoDB for asynchronous document storage of analysis records. Communication between frontend and backend occurs via RESTful API calls with JSON payloads.

B. Analysis Pipeline Design

The analysis pipeline is conceptually organized as five independent modules operating sequentially on the input image:

- Module 1 (Band Detection): Localize all visible protein bands across all lanes, reporting spatial coordinates as resolution-independent percentages.
- Module 2 (Intensity Quantification): Measure relative band intensity on a normalized zero-to-one scale, where one represents the highest-density band.
- Module 3 (Normalization Inference): Identify the normalization strategy implied by the experimental layout, distinguishing housekeeping protein references from total protein stains.
- Module 4 (Uncertainty Estimation): Generate per-band uncertainty estimates expressed as 95% confidence intervals around each intensity measurement.

- Module 5 (Authenticity Scoring): Evaluate membrane texture, band morphology, and background consistency to produce a composite authenticity score.

C. Current vs. Target Implementation

In the current deployed version, all five modules are served by a single structured prompt to the Gemini 1.5 Flash vision model [11]. This approach enables rapid deployment with zero training data requirements. The target research implementation envisions replacing each module with a dedicated neural network: a YOLO-based detector for band localization, a ResNet regression head for intensity measurement, a classification network for normalization inference, Monte Carlo dropout for uncertainty quantification [12], and a patch-based forensic classifier for authenticity detection [10].

IV. IMPLEMENTATION

A. Backend

The FastAPI backend exposes four API endpoints. The primary POST /api/analyze endpoint accepts multipart image uploads (JPEG, PNG, WebP; maximum 10 MB), base64-encodes the image data, constructs a structured multimodal prompt, and dispatches it to the Gemini API. A dedicated JSON parser handles response extraction with multiple fallback strategies to ensure robustness against formatting variations. Analysis records, including the encoded image and complete results, are persisted to MongoDB via the Motor asynchronous driver.

B. Frontend

The React frontend comprises eleven specialized components. The UploadZone component provides drag-and-drop image selection with client-side validation. The BandOverlay component renders detected bands as interactive bounding boxes on the original image. The IntensityChart component (built with Recharts) displays intensity bar charts with confidence interval error bars. The AuthenticityGauge presents a circular score indicator, while the UncertaintyTable provides a detailed per-band statistical breakdown. A JSONExport component enables download of complete machine-readable analysis output.

C. Technology Stack

Component	Technology
Backend	FastAPI (Python 3.11)
Frontend	React.js
Vision AI	Gemini 1.5 Flash
Database	MongoDB (Motor async)
Charts	Recharts
Styling	Tailwind CSS
Server	Uvicorn ASGI

TABLE I: Technology Stack

D. Prompt Engineering

The analysis prompt encodes domain-specific constraints derived from western blot biology. It specifies the expected output JSON schema, instructs the model to report positions as percentages for resolution independence, requests normalized intensity values, and defines the criteria for authenticity assessment including membrane texture analysis, band edge evaluation, and background uniformity checking. This structured prompting strategy effectively converts a general-purpose vision model into a domain-specific analytical instrument.

V. EXPERIMENTAL VALIDATION

System validation was conducted along two axes: software reliability testing and analytical accuracy assessment.

A. Software Testing

Automated tests verified correct API behavior across twelve endpoint tests, eight file validation scenarios, and six JSON parsing edge cases, achieving 100% pass rates across all categories. Integration tests confirmed end-to-end data flow from upload through analysis to database storage and frontend rendering.

B. Analytical Accuracy

Analysis accuracy was evaluated by comparing BlotQuant's automated outputs against manual annotations prepared by trained personnel on a set of thirty western blot images. The system correctly detected and localized clearly visible bands in 87% of test cases. Intensity rankings matched manual assessment in the majority of images. Processing time averaged 12-18 seconds per image versus several minutes for manual analysis.

Metric	BlotQuant	Manual
Processing Time	12–18 s	3–5 min
Band Detection	87%	95% (expert)
Uncertainty Output	Yes (95% CI)	No
Authenticity Check	Yes	No

TABLE II: Automated vs. Manual Analysis Comparison

VI. DISCUSSION

The results demonstrate that a browser-based automated system can deliver comprehensive western blot analysis without requiring any manual parameter selection, addressing a practical barrier that has limited the adoption of quantitative blot analysis in many laboratories. The twelve-to-eighteen second processing time makes the tool suitable for integration into routine experimental workflows.

The current reliance on a general-purpose vision model introduces inherent limitations in detection sensitivity

compared to purpose-trained detectors. Faint bands near the noise floor are occasionally missed, and intensity measurements lack the precision achievable with calibrated optical densitometry. However, the consistency of automated measurements, eliminating the 12% inter-operator variability documented in the literature, represents a meaningful improvement in reproducibility even at the current accuracy level.

The provision of uncertainty estimates, even when derived from model confidence rather than rigorous Bayesian inference, establishes the important practice of reporting measurement intervals rather than bare point estimates. This approach aligns with metrological best practices and directly addresses reproducibility concerns raised by the broader scientific community.

The modular architecture ensures that each analytical component can be independently upgraded as dedicated training data becomes available, without requiring changes to the overall system design or user interface. This evolutionary deployment strategy allows the platform to deliver immediate utility while maintaining a clear development trajectory toward higher analytical rigor.

VII. CONCLUSION

This paper presented BlotQuant, a browser-based platform that automates western blot image analysis through an integrated five-module pipeline encompassing band detection, intensity quantification, normalization inference, uncertainty estimation, and authenticity scoring. The system eliminates manual parameter selection, produces structured machine-readable outputs with confidence intervals, and introduces authenticity verification as a standard component of blot analysis.

Future work will focus on training dedicated deep learning modules for each pipeline stage using weakly supervised labels derived from automated processing of published blot images, implementing Monte Carlo dropout-based uncertainty quantification, developing patch-level forensic classifiers for improved authenticity detection, and extending the platform with batch processing and cross-imager calibration capabilities.

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